

Data Collection and Preprocessing for Contextual Analysis of Aircraft Turnaround Time in Vietnam's Aviation Environment

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Abstract

In the modern aviation ecosystem, understanding the operational dynamics of the supply chain is a vital factor for airline survival. Central to this efficiency is the Aircraft Turnaround Time (TAT). Rather than treating TAT as a pure black-box prediction target, this report focuses on unlocking the causal mechanisms and structural bottlenecks that drive turnaround anomalies. Existing tabular aviation datasets often suffer from context blindness, failing to explain distribution shifts and non-linear disruptions caused by localized environmental shocks. To address this, we introduce Vietnam's Domestic Aviation Multi-Source Dataset, a heterogeneous framework designed specifically for insight-driven analytics. We integrate high-frequency METAR meteorological data with dynamic operational logs, implement a rigorous workflow architecture to eliminate data leakage, and dissect the data through explainable causal analytics. Our empirical findings expose critical non-linear operational behaviors—such as the overnight stop patterns and the cumulative delay propagation paradox in low-cost carriers—proving that structural context is far more critical than naive metric-chasing for real-world aviation decision-making.

Keywords

Aviation operations, Turnaround time, METAR, Data integration, Operational delays.

1 Introduction

In the aviation industry, managing the time an aircraft spends on the ground is highly important for airline efficiency. An aircraft only generates revenue while flying; therefore, any delay during ground operations directly increases operating costs. The central part of this ground cycle is the **Turnaround Time (TAT)**, which is the total time from when an aircraft arrives at the gate until it is ready to depart for the next flight. Understanding the actual causes of turnaround delays is necessary for improving airline schedules and fleet management.

However, analyzing TAT is difficult in Vietnam's aviation environment due to two main challenges. First, as a fast-growing market, Vietnam faces constant problems with **airport infrastructure saturation**. Major domestic hubs often operate at or near their maximum capacity limits. Second, **local weather conditions** frequently disrupt ground activities. These external factors, combined with the tight schedules of low-cost carriers, create non-linear delay propagation across the domestic flight network. This paper aims to analyze these combined operational factors using a clean, data-driven approach.

Research on Vietnam's aviation sector faces a major practical challenge: **the lack of internal operational data**. Crucial information such as airport ground-handling times, detailed maintenance logs, and airline delay records are strictly private. Because of this data scarcity, existing studies often use simple, isolated flight lists. These lists cannot show the actual operational problems in Vietnam, such as airport overloading or weather disruptions.

To solve this practical problem, we collected and combined different public data sources to build **"Vietnam's Domestic Aviation Multi-Source Dataset"**. Instead of focusing only on mathematical predictions, this work aims to explain the real causes of flight delays. This paper has three direct contributions:

- (1) **Data Combination:** We combined public flight tracking data with hourly **METAR** weather reports to analyze how local weather shocks directly affect aircraft turnaround times.
- (2) **Data Organization Method:** We designed an **Operational Workflow Router** to organize these mixed data sources correctly, ensuring that the time sequence of flights is preserved and data errors are avoided.
- (3) **Practical Operational Insights:** Our analysis uncovers real operational patterns specific to the Vietnamese market, such as the constraints of overnight aircraft parking and the cumulative delay patterns in domestic low-cost carrier fleets.

2 Related Work

The analysis of aircraft ground processes and Turnaround Time (TAT) has evolved from rigid deterministic scheduling of passenger operations to complex, multi-source non-linear diagnostics.

2.1 Ground Processes and Critical Path Modeling

Early foundational literature focused heavily on dissecting the sequential nature of ground operations at the gate. Studies by Fricke and Schultz and San et al. pioneered the mathematical modeling of aircraft turnaround behaviors by defining the *Critical Path* of parallel and sequential sub-processes, including cabin cleaning, refueling, and passenger baggage loading. While these traditional operational research frameworks succeeded in quantifying the theoretical baseline of ground-handling efficiency, they operated under highly idealized, isolated assumptions. Consequently, they lack the empirical flexibility to capture dynamic bottleneck propagation when external network disruptions occur.

2.2 Machine Learning and Multi-Source Frameworks

To bridge the gap between theoretical models and real-world volatility, recent methodologies have shifted toward data-driven architectures. A landmark study by Venkatesh et al. pioneered the application of non-linear Machine Learning techniques to prove that TAT is structurally governed by three macro-attribute dimensions:

- (1) **Fleet Architecture:** The physical constraints and size classifications of different aircraft generations (e.g., narrow-body vs. wide-body dynamics).
- (2) **Spatio-Temporal Context:** The localized environmental conditions and real-time high-frequency weather shocks at the time of operation.
- (3) **Infrastructure Saturation Density:** The real-time capacity strain and traffic congestion on runways, aprons, and gates during actual handling windows.

While the multi-source paradigm established by Venkatesh et al. underscores the necessity of heterogeneous data fusion, empirical applications remain heavily biased toward Western aviation ecosystems. There is a distinct lack of open-source frameworks tailored to capture the unique local operational causality and non-linear ground dynamics within highly saturated, infrastructure-constrained regional markets like Vietnam.

3 Dataset and System Architecture

We design a simple and automated data pipeline to handle from data collection to feature engineering across our data sources.

3.1 System Architecture Overview

The pipeline is divided into three straightforward steps:

- **Step 1: Multi-source Data Collection:** The system gathers raw data from different sources, including historical flight logs, METAR weather strings, aircraft characteristics, and airport metadata.
- **Step 2: Spatiotemporal Synchronization:** We use a centralized mapping method to connect all data sources together based on two common keys: the aircraft identifier (tail number) and the operational timestamp (converted to UTC).
- **Step 3: Data Cleaning and Feature Engineering:** The pipeline applies specific aviation business logic to clean the raw logs, remove invalid records, and build the final dataset for our analysis.

3.2 Data Sources and Coverage Details

The dataset strategically covers the Vietnamese domestic market, including the three major airlines: Vietnam Airlines (VNA), Vietjet Air (VJC), and Bamboo Airways (BAV). As shown in Table 1, we focus on the Airbus A320 Family (A320/A321 CEO, NEO, and NEO-ACF) as the backbone of Low-Cost Carrier (LCC) operations, while also covering wide-body aircraft (A330, A350, B787) for distribution diversity.

To automate the data collection pipeline, we compiled a master reference list containing 196 airports operated by the airlines. The primary purpose of this list is to extract the standard IATA and

Table 1: Fleet Composition in the Dataset

Airline	Aircraft Type	Fleet Size	Avg Age
VJC	A320/A321 CEO	24	10.59
	A321 NEO	14	4.19
	A330-300	5	13.55
VNA	A321 CEO	15	14.32
	A321 NEO	11	6.94
	B787/ A350	10	10.33
BAV	A320/A321 CEO	5	12.12

ICAO airport codes. These codes serve as the exact input parameters for our automated API calls, enabling the system to fetch the corresponding high-frequency METAR meteorological strings and real-time airport flight density records.

3.3 Raw Data Collection

The pipeline automates the retrieval of raw data from three primary, independent sources to reconstruct the complete operational context for each flight. The ingestion process focuses on the following components:

- **Historical Flight Logs:** Collected using the specific aircraft registration number (tail number) to reconstruct the sequential itinerary of each aircraft. This continuous sequence preserves the temporal relationship between flights, making it easier to map data and handle missing records effectively.
- **Airport Meteorological Data (METAR):** Fetched using ICAO codes to extract raw weather features (wind, visibility, precipitation).
- **Airport Traffic Density:** Due to hardware and limitations in tracking the entire national airspace simultaneously, the pipeline utilizes the pre-aggregated airport delay and traffic statistics provided directly by FR24.

In total, before any synchronization or filtering, the pipeline successfully collected the raw data streams as summarized in Table 2. This raw dataset establishes the baseline for subsequent data cleaning and spatiotemporal mapping.

Table 2: Summary of Collected Raw Data Streams

Data Stream	Primary Source	Total Records
Flight Lists	Flightradar24	17,397
Airport Stats	Flightradar24 (Airport)	78,658
Weather Stats	Aviation Weather Center	324,558

4 Data Preprocessing and Cleaning

Raw data ingested from distributed APIs contains systemic biases due to local ADS-B receiver limitations or synchronization delays in the Flightradar24 platform. To reconstruct dataset, we implement two procedures: domain-driven imputation and temporal standardization.

4.1 Domain-Driven Missing Data Imputation

Instead of employing naive deletion, which leads to significant sample loss, we leverage the physical constraints of the flight network to impute missing values:

- **Actual Time of Arrival (ATA) Inference:** For records with valid Departure times (*ATD*) but missing *ATA* due to signal loss during the final approach, the system applies a regional scanning module. The *ATA* is estimated as:

$$ATA_{\text{imputed}} = ATD + \overline{\Delta T_{\text{flight}}} \quad (1)$$

where $\overline{\Delta T_{\text{flight}}}$ is the historical average flight duration for the specific origin-destination pair. Such records are flagged as *Estimated* to maintain transparency for downstream modeling.

- **Origin/Destination Recovery:** If the FROM or TO field is missing, the system performs a recursive sequence check over the chronological flight history of the same aircraft registration. Since consecutive legs of the same aircraft must satisfy itinerary continuity, the missing destination of flight i can be inferred from the known origin of flight $i + 1$, and the missing origin of flight i can be inferred from the known destination of flight $i - 1$. For domestic round-trip patterns, the system additionally checks whether two adjacent legs form a plausible outbound–return pair between the same two airports. A recovery is accepted only when the inferred airport is consistent with the surrounding tail-number sequence and the temporal gap remains within the operational turnaround window; otherwise, the record is left removed.
- **Critical System Noise Isolation:** Records suffering from simultaneous loss of *ATD* and *ATA*, or those resulting from long-duration "blind spots" (exceeding 12 hours), are isolated and pruned to protect the model from extreme stochastic noise.

4.2 Temporal Standardization and Timezone Alignment

A primary challenge in multi-source aviation data is timezone conflict. To ensure mathematical consistency, all timestamps (Local Time from various airports) are unified into **UTC (Greenwich Mean Time - GMT 0)**:

- **Operational Status Logs:** Local landing times are mapped against an $GMT_Timezones$ to perform algebraic subtraction (e.g., $UTC = Local - 7$ for Vietnam).
- **Static System Collection:** Since airport congestion metrics are captured via local machine clocks, an inverse transformation is applied to align these snapshots with the global flight timeline, enabling a *Spatiotemporal Join*.

5 Feature Engineering

5.1 Flight Feature Engineering and Target Generation

The pipeline processes the sequential flight logs of each aircraft to extract internal operational features and the target variable:

- **Sequence Reset (> 240 mins):** If ground-time between consecutive legs exceeds 240 minutes, the system identifies it as a night stop or maintenance window, automatically resetting and starting a new operational sequence.
- **Cumulative Delay:** Tracks the total running delay accumulated within each active operational sequence to capture how previous delays affect the current turnaround.
- **Target Variable (Actual TAT):** Calculated directly as the exact temporal difference between the actual arrival time (*ATA*) of the current flight and the actual departure time (*ATD*) of the next flight leg.

5.2 Meteorological Feature Engineering via METAR Parsing

As illustrated in the example in Figure 1, the raw METAR strings collected from the stations cannot be used directly and must be parsed into structured features. We designed an automated extraction component based on *Regular Expressions (Regex)* to decode these text strings into 6 core contextual variables:

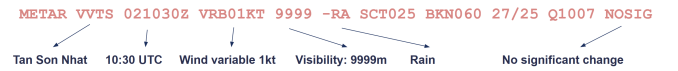


Figure 1: An example of a METAR report and its decoded variables.

- **is_cavok:** Binary flag for ideal ceiling and visibility.
- **visibility_meters:** Runway's visibility.
- **wind_speed & is_gust:** Surface wind dynamics affecting runway approach.
- **has_thunderstorm & has_heavyrain:** Extreme weather indicator that triggers ground stops.
- **has_low_vis_fog:** Common in Northern Vietnam.

6 Data Merging

To prevent **Data Leakage**, we implement a **Time-Bounded Backward Join**. For every flight record i with a Scheduled Time of Departure (STD_i), the system queries the environmental state:

$$\{T_{\text{weather}}, T_{\text{airport}}\} \leq STD_i \quad (2)$$

The optimal contextual snapshot $\hat{T}_{\text{context}}$ is selected using a nearest-neighbor approach:

$$\hat{T}_{\text{context}} = \arg \min_{T \leq STD_i} (STD_i - T) \quad (3)$$

This ensures the model only accesses information available at the moment the turnaround decision is made at the Airport Operational Control Center (AOCC).

7 Data Validation

After combining the flight logs, METAR data, and airport metrics based on time and location, we checked the dataset carefully. This step ensures that the flight timeline is logical and correct, providing a clean dataset for analyzing.

7.1 Post-Merge Missing Value Diagnostics

A systematic diagnostics of the consolidated dataset revealed distinct, non-random missingness patterns. As quantified in Table 3, these gaps correlate directly with the operational characteristics of their primary ingestion sources rather than stochastic system failures.

Table 3: Missing Value Distribution in Consolidated Dataset

Feature Group	Missing Count	Missing Rate (%)
METAR Variables	1,192	7.99
Airport Delay Indices	209	1.40
Scheduled Turnaround	81	0.54
Core Flight Identifiers	0	0.00

- **Structural Gaps in Meteorological Features:** Variables decoded via regular expressions from the METAR strings (*wind_speed*, *is_cavok*, *has_thunderstorm*) exhibit a synchronized missing rate of approximately 7.99% (1,192 records). This missingness is heavily concentrated in flight legs operating at smaller domestic regional airports within Vietnam. While major international hubs (e.g., Noi Bai - VVNB, Tan Son Nhat - VVTS) consistently broadcast high-frequency METAR packets, smaller domestic nodes lack the automated telemetry infrastructure to publish regular weather summaries to global repositories, creating data blocks.
- **Non-scheduled Flights Data Gap:** Some flights (such as charter or non-scheduled operations) do not have a pre-planned schedule in the system before data collection. This creates missing values in the airport delay metrics.
- **Baseline Integrity:** Core operational flight identifiers (*DATE*, *FLIGHT*, *FROM*) maintain a 0.0% missing rate, validating the foundational robustness of our aircraft-centric base table.

7.2 Handling Missing Weather Data

In our dataset, missing weather records do not occur randomly or temporarily. Instead, certain regional airports in Vietnam do not broadcast public **METAR** reports at all. Simply deleting all flight rows that lack weather data would create a heavily biased dataset, leaving only major international hubs.

To keep the data honest and realistic, we applied a straightforward rule: for flights operating at regional airports with no METAR broadcasting, the weather features are strictly left as *Null*. We do not use any statistical guessing or artificial imputation here. This approach ensures that our analysis of operational delays (Δ) reflects the actual infrastructure limits of each local airport without introducing artificial noise.

7.3 Outlier Filtering and Operational Logic Bounds

An analysis of the collected Actual Turnaround Time (Actual TAT) showed extreme errors at both ends of the data distribution. The raw values ranged from a minimum of -1888.0 minutes to a maximum of 30116.0 minutes. These impossible numbers come from

telemetry logging errors in the Flightradar24 data stream or aircraft undergoing long-term maintenance.

To clean the dataset without losing real operational information, we applied two strict rules based on civil aviation practices:

- **Lower Bound (< 30 minutes):** We removed records with a TAT below 30 minutes, including all negative values. Under domestic ground-handling limits, a commercial jet requires at least 30 minutes for basic passenger deboarding, cleaning, and re-boarding.
- **Upper Bound (> 1440 minutes):** We removed records where ground time is longer than 1 day. Ground times longer than that usually mean the aircraft is stopped for technical repairs, which are not part of normal flight turnarounds.

8 Final Dataset Profile and Characteristics

Following the enforcement of structural missingness filters and domain-driven operational logic bounds, the final validated dataset was locked for down-stream modeling tasks.

The clean, post-processed dataset comprises a total of 17,107 valid flight transactions, with each record spanning across 20 continuous and categorical variables (including 19 input parameters and 1 target variable). A comprehensive schema defining the data types, descriptions, and operational examples for every feature within this consolidated matrix is detailed in Table 4. This balanced and clean feature space ensures that subsequent predictive models can learn legitimate operational patterns without being skewed by data collection artifacts.

9 Exploratory Data Analysis (EDA)

In this section, we conduct an Exploratory Data Analysis (EDA) on the locked 17,107 flight records to uncover the underlying empirical patterns and relational dependencies within the aviation network. Rather than treating the features as isolated statistical inputs, the objective of this diagnostic phase is to map out how the temporal dynamics, aircraft scheduling profiles, and localized meteorological variations interact with the ground handling timeline.

Specifically, our analysis focuses on three core investigations: (1) isolating the distributional properties of the actual turnaround time, (2) tracking the propagation of cumulative delays across consecutive flight legs, and (3) assessing the operational degradation caused by critical weather triggers parsed from the METAR streams. This systematic baseline profiling ensures that the downstream predictive models are aligned with genuine operational behaviors.

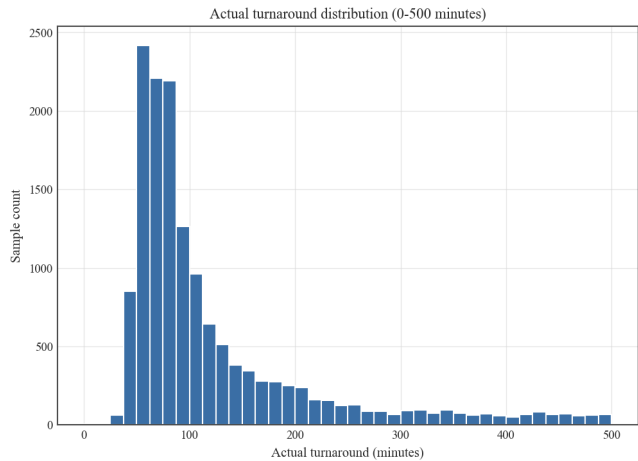
9.1 Target Variable Distribution Analysis

To evaluate the baseline properties of the target variable, we analyze its empirical distribution and percentile growth metrics prior to the truncation phase.

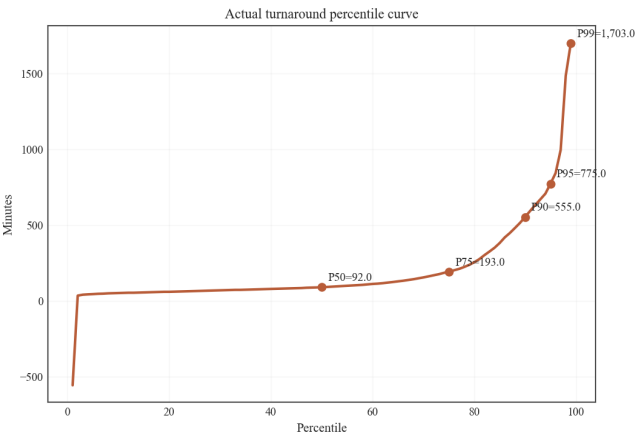
As illustrated in Figure 2a, the target variable exhibits a prominent right-skewed distribution, with the core operational density packed within the 50–90 minute interval and peaking between 50 and 75 minutes. Beyond the 100-minute mark, the sample count decays exponentially into a long continuous tail. This structural skewness is quantified by the percentile trajectory in Figure 2b: the lower tail plunges to a minimum of -1703.0 minutes due to logging artifacts, while the operational median (P_{50}) sits at 92.0 minutes.

Table 4: Feature Dictionary of the Final Dataset

Feature Name	Data Type	Description / Operational Example
<i>Aircraft Profile Features</i>		
reg_number	String	Tail registration number of the aircraft (e.g., VN-A123, EU0005).
aircraft_model	String	Specific aircraft variant model designation (e.g., A320-211, B787-9).
age	Integer	Operational age of the airframe since manufacturing date (years).
airline_code	String	ICAO/IATA operating airline code (e.g., VNA, VJC).
<i>Flight Itinerary Features</i>		
FLIGHT	String	Unique commercial flight number designation (e.g., VJ123, QH345).
FROM	String	IATA code of the departure origin airport (e.g., SGN).
TO	String	IATA code of the arrival destination airport (e.g., HAN).
std_utc	SmallDateTime	Scheduled Time of Departure in Coordinated Universal Time (UTC).
scheduled_turnaround	Integer	Baseline planned turnaround buffer allotted by airline schedule (minutes).
flight_legs	Integer	Sequential position index of the leg within the daily aircraft rotation.
cumulative_delay	Integer	Total accrued delay minutes of the aircraft prior to the current leg.
<i>Meteorological Features (METAR)</i>		
is_cavok	Boolean	Ceiling and Visibility OK binary indicator flag (1 = Ideal weather).
wind_speed	Float	Surface wind speed parsed from localized METAR telemetry (knots).
is_gust	Boolean	Wind gust activity indicator flag at the station.
visibility_meters	Integer	Runway’s visibility at the airport.
has_thunderstorm	Boolean	Convective thunderstorm activity indicator flag within terminal area.
has_heavy_rain	Boolean	Heavy precipitation / rain downpour binary indicator flag.
has_low_vis_fog	Boolean	Low visibility/fog conditions affecting runway approach visibility.
<i>Airport Ground Saturation Metrics</i>		
dep_recent_index	Float	Rolling ground handling operational index provided by FR24 telemetry.
dep_recent_delay_avg	Integer	Pre-aggregated rolling average of departure delay occurrences at the node.
<i>Target Variable</i>		
actual_turnaround	Integer	Measured actual turnaround time on ground (minutes) [Target Variable].



(a) Frequency distribution (0–500 minutes window)



(b) Actual turnaround percentile curve

Figure 2: Empirical distribution diagnostics of the raw Actual TAT target variable.

Past P_{75} (193.0 minutes), the curve scales non-linearly through P_{90} (555.0 minutes) and P_{95} (775.0 minutes). This quantitative profiling

proves that the extreme tails are heavily contaminated by tracking anomalies, fully justifying the necessity of the 20–240 minute operational logic truncation.

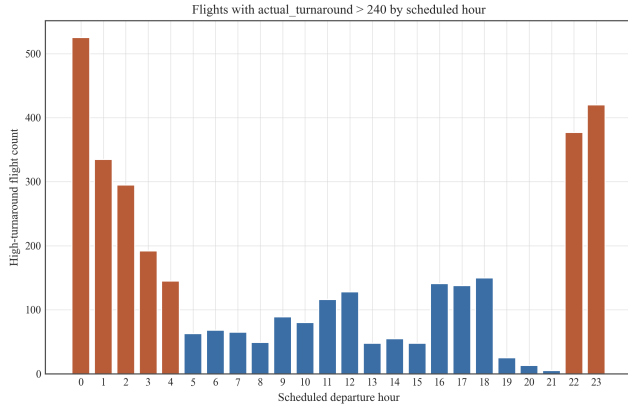


Figure 3: Analysis of extended turnaround times (> 240 minutes) by scheduled departure hour.

9.2 Temporal Analysis of Extended Ground Durations

To validate the operational nature of durations exceeding the 240-minute threshold, we analyze their temporal distribution across scheduled departure hours in Figure 3.

As demonstrated in Figure 3, the occurrences of extended ground times are strictly concentrated within the late-night and early-morning windows (22:00 to 04:00), highlighted by the orange clusters. While the absolute count peaks during midnight slots, the internal rate within each hour surges drastically to 60%–75% between 22:00 and 23:00 UTC. This empirical pattern confirms that these prolonged periods do not represent active commercial turnarounds; instead, they capture systemic *Night Stops* (overnight positioning) and scheduled overnight line maintenance. This temporal behavior validates our domain-driven truncation strategy, as embedding these stationary non-operational states would artificially distort the model’s ability to predict daytime flight efficiency.

9.3 Contextual and Institutional Influences on Turnaround Efficiency

9.3.1 Airline Carrier Profiles and Scheduling Strategies. Table 5 quantifies the operational relationship between planned ground buffers (*scheduled_turnaround*) and on-time departure rates ($|actual - scheduled| \leq 15$ minutes).

The empirical data reveals a structural divergence between low-cost and legacy carrier models. VietjetAir (VJC) maintains a highly conservative baseline buffer, yet correlates with the lowest punctuality vector. Conversely, Vietnam Airlines (VNA) and Bamboo Airways (BAV) operate under tighter ground constraints but preserve significantly higher schedule integrity.

This technical trade-off demonstrates that extending the scheduled turnaround window is insufficient to mitigate delay propagation. The lower punctuality in the LCC model (VJC) is driven by compounding rotational delays across sequential flight legs and high airframe utilization, whereas legacy networks optimize slot predictability to secure stable ground handling timelines despite compressed windows.

Table 5: Airline Operational Policies and On-Time Performance (< 240 Minutes)

Airline Code	Sched. TAT (min)	On-Time Rate (%)
VietjetAir (VJC)	68.4	37.0%
Vietnam Airlines (VNA)	62.6	51.4%
Bamboo Airways (BAV)	51.1	50.4%

9.3.2 Airport Congestion Saturation and Turnaround Deviations. Figure 4 benchmarks the temporal alignment between localized airport congestion metrics (*Median dep_recent_index*) and the corresponding volume of anomalous ground handling deviations ($|actual - scheduled| > 15$ minutes) across major domestic hubs.

The empirical curves demonstrate that there is no strong, continuous linear correlation between standard baseline airport density and operational turnaround spikes. High congestion indices do not mechanically trigger a proportional surge in ground delays across all periods.

Instead, the relationship is driven by non-linear capacity shocks and hub-specific operational constraints. Significant deviation peaks are heavily concentrated at primary hubs like Tan Son Nhat (SGN) and Noi Bai (HAN) during specific operating windows. This indicates that while nominal traffic density is generally absorbed by standard airport buffers, localized bottlenecks and sudden operational saturation at saturated hubs act as stochastic triggers that disrupt ground handling predictability.

9.3.3 Conclusion of EDA and Structural Implications. The preliminary exploratory data analysis transitions the analytical focus from basic predictive mapping to understanding complex, non-linear system behaviors. The structural insights extracted from the target variable distribution, temporal carrier profiles, and airport congestion indices establish two definitive conclusions regarding the nature of the data:

First, **strong non-linearity dominates the operational ecosystem**. The relationships between the ground handling deviations and external features do not follow predictable linear trajectories. Instead, they operate under threshold-driven capacity shocks. This is evident where an increase in scheduled ground buffers correlates paradoxically with lower punctuality, and where airport density triggers severe turnaround anomalies only when hitting localized saturation ceilings at major hubs (SGN, HAN).

Second, the structural variance in the data is fundamentally governed by unobserved, non-linear network dynamics—specifically, asymmetric rotational delay cascades across tightly chained flight legs.

Consequently, traditional linear or additive statistical models (e.g., OLS Regression) are mathematically insufficient to capture these boundary conditions and phase transitions. This empirical reality fully validates the strategic necessity of utilizing robust machine learning architectures—such as gradient-boosted decision trees (XGBoost, LightGBM) or non-linear neural networks.

9.4 Data Limitations and Constraints

While our data integration provides useful insights, this study operates under three practical constraints:

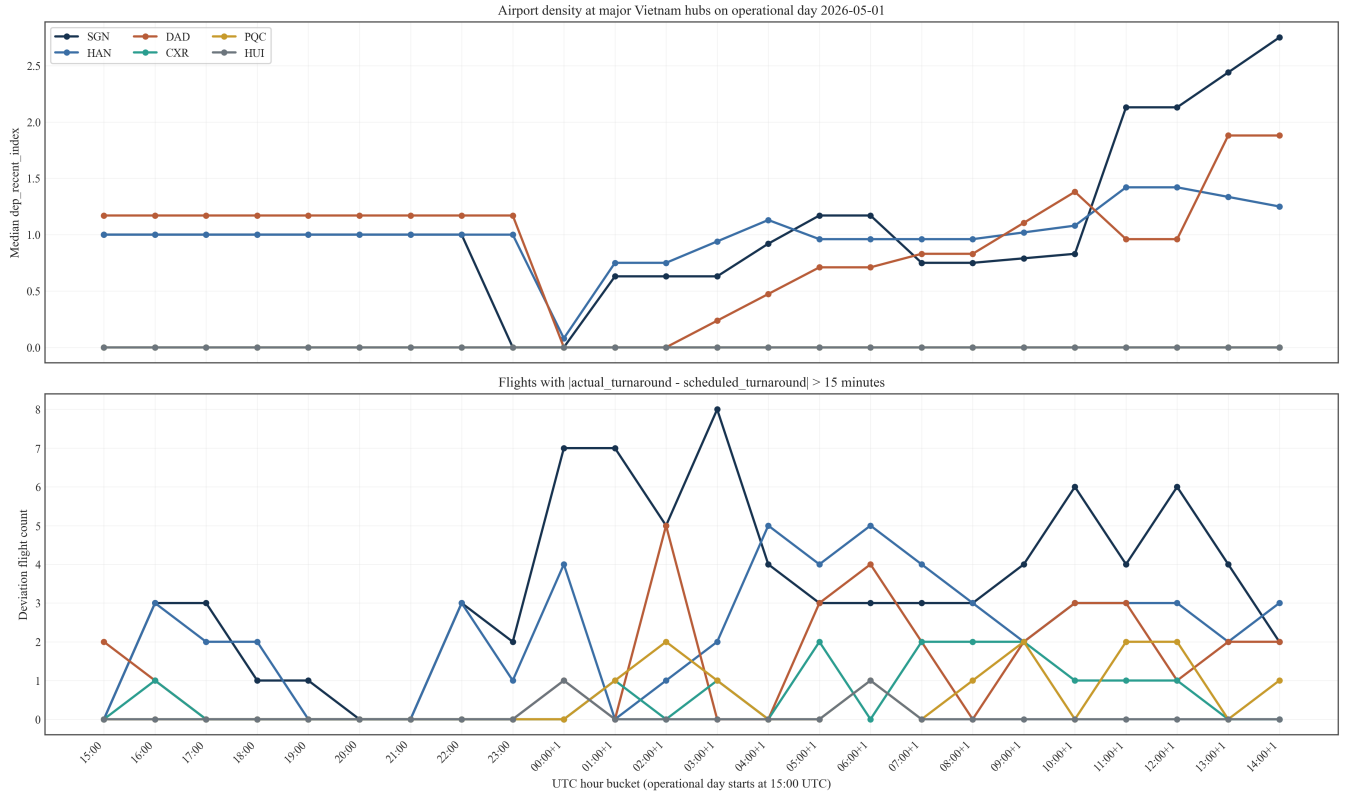


Figure 4: Temporal comparison between airport departure density (top) and significant turnaround time deviations (bottom) across domestic hubs.

- **Data Duration:** Due to data access policies and historical logging limits of third-party streams, our dataset covers a limited time window. This prevents us from analyzing long-term seasonal trends or multi-year policy changes.
- **Lack of Internal Metrics:** Our data views airline operations from an external perspective. It lacks internal airline metrics that directly impact ground handling, such as crew schedules, real-time maintenance logs, and exact timestamps for refueling, catering, or baggage loading.
- **Complex Feature Interactions:** Ground operations are highly interconnected, where a delay in one step cascades into others. Because these real-world variables interact in a non-linear way, isolating the exact causal impact of a single factor on the deviation (Δ) is highly difficult.

10 Conclusion and Future Work

This report introduced Vietnam's Domestic Aviation Multi-Source Dataset, a heterogeneous framework designed to mitigate context blindness in traditional aviation analytics by integrating dynamic operational logs with high-frequency METAR meteorological data. Through a rigorous, temporal leakage-aware pipeline, we have successfully synchronized distributed tracking and environmental variables while preserving operational domain logic. Our exploratory

data analysis (EDA) validated the integrity of the dataset, uncovering critical non-linear operational behaviors, including structural overnight stop characteristics and systemic delay propagation patterns within low-cost carrier rotations. While this study establishes a robust data foundation and delivers initial diagnostic insights, it lays the groundwork for advanced predictive modeling. Future research will leverage this unified dataset to train non-linear machine learning architectures and deep learning frameworks for highly accurate, context-aware aircraft turnaround time (TAT) estimation and multi-hop delay cascade forecasting.

References

- [1] H. Fricke and M. Schultz. 2009. Tactical aircraft turnaround simulation—A competitive tool for airlines and airports. In *Proceedings of the Winter Simulation Conference (WSC)*. 2801–2812.
- [2] B. San, M. Schultz, and H. Fricke. 2016. Aircraft turnaround process modeling under idealized and operational constraints. *Journal of Air Transport Management* 55 (2016), 120–132.
- [3] N. Venkatesh, B. Gopalakrishnan, and G. Keller. 2020. A data-driven machine learning framework for aircraft turnaround time estimation considering spatio-temporal context and infrastructure saturation. *Transportation Research Part C: Emerging Technologies* 119 (2020), 102745.
- [4] Flightradar24. 2026. *Aviation Flight Tracking Data and Historical Aircraft Turnaround Records*. Retrieved May 1, 2026 from <https://www.flightradar24.com>
- [5] Aviation Weather Center (AWC). 2026. *Historical Aircraft Meteorological Data and High-Frequency Weather Observations*. National Oceanic and Atmospheric Administration (NOAA). Retrieved May 1, 2026 from <https://www.aviationweather.gov>

A Datasheet for Dataset (Gebru et al., 2021)

This appendix summarizes the dataset documentation following the spirit of the standardized *Datasheets for Datasets* framework proposed by Gebru et al. (2021). The purpose is to improve transparency regarding dataset motivation, composition, collection provenance, preprocessing decisions, limitations, and responsible use.

A.1 Motivation

Q: For what purpose was the dataset created?

A: The dataset was created to analyze and profile the systemic, non-linear structural behaviors of aircraft turnaround operations in Vietnam’s domestic aviation environment. Rather than treating turnaround time as a simple black-box prediction target, the dataset aims to capture the operational context behind turnaround anomalies, including aircraft rotation history, airport congestion, and local meteorological conditions.

Q: Why was a new dataset needed?

A: Public aviation datasets often provide isolated flight records without sufficient contextual information. In Vietnam’s aviation environment, turnaround behavior is strongly affected by airport saturation, weather shocks, airline scheduling policies, and aircraft rotation chains. Therefore, a multi-source dataset was required to reconstruct the spatiotemporal context surrounding each turnaround event.

Q: Who created the dataset and who funded its creation?

A: The dataset was curated independently by the authors for academic study. No external commercial funding was involved.

A.2 Composition

Q: What do the instances that comprise the dataset represent?

A: Each instance represents one reconstructed domestic commercial aircraft turnaround operation, defined as the ground interval between the actual arrival of one flight leg and the actual departure of the subsequent flight leg operated by the same aircraft registration. Each record contains aircraft profile attributes, flight itinerary information, temporal variables, parsed METAR weather indicators, airport ground saturation metrics, and the target variable `actual_turnaround`.

Q: What is the total sample size and coverage?

A: The final validated dataset contains $n = 17,107$ valid turnaround operations after cleaning and filtering. The dataset covers aircraft operated by Vietjet Air (VJC), Vietnam Airlines (VNA), and Bamboo Airways (BAV). The master reference registry includes 84 aircraft registrations and 196 airports with corresponding IATA/ICAO mappings.

Q: Does the dataset contain all possible instances or is it a sample?

A: It is a filtered observational sample collected from public aviation data streams. The dataset does not contain all domestic flights in Vietnam. It is restricted to flights that could be consistently reconstructed from available aircraft histories, airport metadata, weather records, and operational timestamps within the selected collection period.

A.3 Collection Process

Q: How was the data associated with each instance acquired?

A: The dataset was acquired through an incremental multi-source data collection pipeline. A master reference registry was first constructed to store aircraft registration numbers, airline identifiers, aircraft metadata, and airport IATA/ICAO codes. From this registry, source-specific collectors generated API calls for three major data streams: historical aircraft flight logs from Flightradar24, METAR weather reports from aviation weather archives, and airport-level traffic and delay indicators from Flightradar24 airport statistics.

Q: Was the dataset collected through a one-shot download?

A: No. The dataset was collected through daily and source-specific acquisition cycles because public flight tracking platforms do not expose a complete historical dump for all aircraft and airports. Raw responses from each source were stored as separated historical files before synchronization, cleaning, and merging. This design preserves provenance and allows downstream preprocessing steps to be audited.

A.4 Preprocessing, Cleaning, and Missingness

Q: Was any preprocessing, cleaning, or labeling done?

A: Yes. The preprocessing pipeline includes timestamp standardization, aircraft sequence reconstruction, missing value handling, outlier filtering, METAR parsing, feature engineering, and time-bounded data merging. All timestamps were converted to UTC to avoid timezone conflicts across airports and data sources. Contextual weather and airport-density features were merged using a backward temporal join, ensuring that only information available before the scheduled departure time was attached to each flight record.

Q: How were missing values handled?

A: Missing values were handled according to their data-generating mechanism. Missing destination fields could be recovered when the subsequent flight origin of the same aircraft provided sufficient evidence. Partial missing arrival or departure timestamps were reconstructed only when the route history and surrounding sequence made the inference operationally plausible. In contrast, missing METAR values at regional airports were not statistically imputed, because such gaps reflect structural limitations in public weather reporting infrastructure rather than random noise.

Q: How were outliers handled?

A: Unrealistic or non-operational turnaround values were filtered using aviation-domain logic. Negative turnaround times and extremely short ground intervals were treated as logging artifacts, while very long ground intervals were considered likely night stops, maintenance windows, or inactive aircraft states rather than normal commercial turnarounds. These rules were applied to isolate active turnaround operations from structural anomalies.

Q: What internal or “dark” data is missing?

A: The dataset does not contain airline-internal operational data, including crew schedules, gate assignments, maintenance logs, refueling milestones, catering progress, baggage loading timestamps, or official airline delay codes. Therefore, the dataset should be interpreted as an external operational reconstruction rather than a complete internal view of ground handling.

A.5 Uses, Limitations, and Distribution

Q: Has the dataset been used for any tasks already?

A: In this study, the dataset was used for exploratory data analysis, structural feature profiling, turnaround-time distribution analysis, and the examination of delay propagation patterns across aircraft rotations.

Q: What are the recommended uses of the dataset?

A: The dataset is suitable for analyzing aircraft turnaround behavior, airport congestion effects, weather-aware operational patterns, delay propagation, and baseline machine learning experiments for turnaround-time estimation. It is also useful as a case study for multi-source aviation data preprocessing and temporal leakage-aware data integration.

Q: What are the non-recommended uses of the dataset?

A: The dataset should not be used to make strong causal claims about airline performance, ground-handling quality, or airport management efficiency without additional internal operational data. Observed patterns should be interpreted as external signals derived from public data sources, not as complete causal explanations.

Q: Will the dataset be publicly distributed?

A: No. Due to upstream data-access constraints and redistribution limitations associated with the primary tracking providers, the full raw dataset is not intended for unrestricted public release. However, the feature schema, preprocessing rules, aggregate statistics, and selected anonymized examples may be shared where permitted for academic verification and reproducibility.